

5. There are a number of factors that should be taken into consideration when deciding what makes the 'best fuel'.

Information was collected about various factors for three fuels, **A**, **B** and **C**.

Fuel A

- Existing supplies will last around 50 years
- Releases 2.8 kJ of energy per gram of fuel burned
- Costs 0.03p per gram of fuel burned
- Burns very easily and no storage issues
- Releases carbon dioxide and water vapour when it burns

Fuel B

- There are infinite supplies of this fuel
- Releases 44.1 kJ of energy per gram of fuel burned
- Costs 0.18p per gram of fuel burned
- Burns very easily but can be difficult to store
- Releases water vapour when it burns

Fuel C

- Existing supplies will last around 250 years
- Releases 1.2 kJ of energy per gram of fuel burned
- Costs 0.04p per gram of fuel burned
- Burns very easily and fairly easy to store
- Releases carbon dioxide, sulfur dioxide and water vapour when it burns

This information was analysed by a group of students to decide what they considered to be the 'best fuel'.



- (a) Give the reason why the information about how easily each fuel burns was not useful to the students when reaching their decision. [1]

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- (b) One of the students based his decision purely on a judgement of the available supply of each fuel. Choose the order that shows his conclusion. Place a tick (✓) in the appropriate box. [1]

best fuel	A	A	B	B	C	C
	B	C	A	C	A	B
	C	B	C	A	B	A
	☐	☐	☐	☐	☐	☐

- (c) Which of the following statements best describes how the fuels affect the environment when they burn? Tick (✓) the correct answer and give the reason for your choice. [2]

all of the fuels contribute to acid rain and global warming when they burn

☐

fuels **A** and **C** contribute to acid rain and global warming when they burn

☐

only fuel **C** contributes to acid rain and global warming when it burns

☐

none of the fuels contribute to acid rain and global warming when they burn

☐

Reason

.....



(d) The cost efficiency of fuel **A** can be calculated as follows:

cost efficiency = $\frac{2.8}{0.03}$ = 93.3 kJ/p

Use the information given for fuel **B** and this example to calculate the cost efficiency of fuel **B**. [2]

Cost efficiency = kJ/p

(e) The students eventually agreed on the following rank order for the fuels.



In the table below, tick (✓) **all** the statements that are **correct** and could therefore have been used in deciding upon this order. [2]

	(✓)
fuel C will run out after fuels A and B	
fuel C is easier to store than fuel A	
fuel A burns more easily than fuel C	
fuel B is the cleanest fuel	
fuel B is easier to store than fuel C	
fuel B will never run out	
fuel A is less harmful to the environment than fuel C	
fuel A is less cost efficient than fuel B	

